

OPERATING SYSTEM

Basics, User Interface & Common Utilities

Introduction to Operating Systems

Files, Folders, Applications and Utilities

Objectives of this unit

- **Understand the concept of an Operating System**
- **Identify popular Operating Systems**
- **Use the mouse, icons, menus and windows**
- **Work with files, folders and directories**
- **Create shortcuts and use common utilities**

Meaning of Operating System

- **System software that acts as an interface between the user and computer hardware**
- **Manages CPU, memory, storage and input/output devices**
- **Provides a platform for running applications**
- **Examples: Windows, Linux, macOS, Android and iOS**

OBJECTIVES OF OPERATING SYSTEM

Operating System (OS) is system software that controls and coordinates the use of hardware among different application software and users. OS acts as an intermediary between the computer user and computer hardware.

The user gives a command, and the OS translates the command into a form that the machine can understand and execute.

Main Objectives of OS

Convenience: To make the computer system convenient and easy to use for the user.

Efficiency: To use computer hardware in an efficient way by handling the details of hardware operations.

OS Provides Convenience to the User

- **OS hides the working of hardware from the user and makes it convenient to use the machine.**
- **Application programs require hardware resources during processing.**
- **Examples include:**
 - **Displaying an application's user interface.**
 - **Loading a program into memory.**
 - **Using I/O devices.**
 - **Allocating CPU to different processes during execution.**

- **Storing or loading data from the hard disk.**
- **The user gives commands to the OS, and the OS handles all the operational steps.**
- **The user need not know how these actions are performed.**
- **OS provides an interface for application programs to interact with hardware.**

OS Manages Computer Resources

- **The different resources of computer hardware have to be managed and controlled by the OS. This includes:**
 - **Managing communication between different devices.**
 - **Controlling the sequence and execution of processes.**
 - **Allocating space on the hard disk.**

- **Providing error-handling procedures.**
- **Supervising and managing the hardware of the computer.**
- **Commonly Used Operating Systems: Microsoft Disk Operating System (MS-DOS), Windows, Windows XP, Linux, UNIXMac**

Types of OS

OS are classified into different types depending on their capability of processing:

- **Single User and Single Task OS**
- **Single User and Multitasking OS**
- **Multitasking OS**
- **Multiprocessing OS**
- **Real Time OS**

Functions of an Operating System

- **Process Management – manages running programs**
- **Memory Management – manages RAM**
- **File Management – organizes files and folders**
- **Device Management – controls hardware devices**
- **Security Management – protects data and resources**
- **User Interface – enables interaction with the computer**

Basics of Popular Operating Systems

- **Microsoft Windows – user-friendly GUI and widely used on PCs**
- **Linux – open-source OS with many distributions such as Ubuntu**
- **macOS – developed by Apple for Mac computers**
- **Android – mainly used in smartphones and tablets**
- **iOS – developed by Apple for iPhone and related devices**

The User Interface

- **User Interface (UI) allows users to interact with the computer**
- **Graphical User Interface (GUI): windows, icons, menus and buttons**
- **Command-Line Interface (CLI): interaction using text commands**
- **Examples: Windows GUI and Linux Terminal**

Desktop and Common Elements

- **Desktop background**
- **Icons**
- **Taskbar / Dock**
- **Start / Menu button**
- **Notification area / System tray**
- **Recycle Bin / Trash**

Mouse Use

- **Move** – moves the pointer
- **Single Click** – selects an item
- **Double Click** – opens an item
- **Right Click** – displays a context menu
- **Drag and Drop** – moves an item
- **Scroll** – moves through a document or webpage

Using the Right Mouse Button

- **Right-clicking an item displays a Context Menu**
- **Common options: Open, Copy, Cut, Paste, Rename, Delete and Properties**
- **Available options depend on the selected item**

Moving Icons on the Screen

- **Place the pointer on the icon**
- **Press and hold the left mouse button**
- **Drag the icon to the desired location**
- **Release the mouse button**
- **This action is called Drag and Drop**

Common Icons

- **This PC – access drives and storage**
- **Recycle Bin / Trash – stores deleted items temporarily**
- **Folder – stores files and other folders**
- **Browser – accesses the Internet**
- **Settings – changes system settings**
- **Search – searches files and applications**

Status Bar

- **Displays information about the current application or window**
- **May show number of items, current page, file size or selection information**
- **May also show zoom level and operation status**

Menus and Menu Selection

- **A menu contains a list of commands or options**
- **Common menus: File, Edit, View, Insert and Help**
- **Click the menu and select the required command**
- **The selected command is then executed**

Running an Application

- **Double-click its desktop icon**
- **Select it from the Start / Applications menu**
- **Use the Search option**
- **Open an associated file**
- **Example: Start → All Apps → Microsoft Word**

Files, Folders and Directories

- **File – stores information or data, e.g. .docx, .pdf, .jpg and .txt**
- **Folder – organizes files and can contain subfolders**
- **Directory – another common term for a folder, especially in Linux**
- **Example: Documents → Computer → Operating System.docx**

Creating a File and Folder

- **Create a folder: Right-click → New → Folder → enter name → Enter**
- **Create a file: Open an application → create content → Save**
- **Choose the required file name and storage location**

Renaming Files and Folders

- **Select the file or folder**
- **Right-click and select Rename**
- **Type the new name**
- **Press Enter**
- **Example: Computer_Basics.docx → Operating_System.docx**

Opening and Closing Windows

- **Minimize** – hides the window temporarily
- **Maximize** – enlarges the window
- **Restore** – returns to the previous size
- **Close (X)** – closes the window
- **Alt + F4** – closes the active window

Using Help

- **Provides information about an application or operating system**
- **Find instructions and understand features**
- **Troubleshoot common problems**
- **Search for commands and keyboard shortcuts**
- **F1 opens Help in many applications**

Creating Shortcuts

- **A shortcut provides quick access to a file, folder or application**
- **Saves time and provides quick access**
- **Useful for frequently used applications**
- **Deleting a shortcut normally does not delete the original file or application**

Basics of OS Setup

- **Start the installation**
- **Select language and region**
- **Select installation location**
- **Create or configure a user account**
- **Set password and network options**
- **Install device drivers and system updates**

Common OS Utilities

- **File Manager** – manages files and folders
- **Calculator** – performs calculations
- **Notepad / Text Editor** – creates simple text files
- **Disk Management** – manages storage devices
- **System Settings** – changes system configuration
- **Task Manager / System Monitor** – monitors processes
- **Backup Utility** – helps protect data
- **Search Utility** – finds files and applications

Important Keyboard Shortcuts

- **Ctrl + C – Copy**
- **Ctrl + X – Cut**
- **Ctrl + V – Paste**
- **Ctrl + Z – Undo**
- **Ctrl + S – Save**
- **Ctrl + A – Select All**
- **Alt + Tab – Switch applications**
- **Alt + F4 – Close active window**
- **Win + E – Open File Explorer**
- **Win + D – Show Desktop**

Summary

- **An Operating System manages computer hardware and software**
- **Windows, Linux, macOS, Android and iOS are popular operating systems**
- **GUI uses icons, menus, windows and buttons**
- **Files and folders organize information**
- **Shortcuts provide quick access**
- **OS utilities help manage and maintain the computer**

Quick Assessment

- **What is an Operating System?**
- **List any four functions of an OS**
- **What is the difference between single-click and double-click?**
- **What is a context menu?**
- **What is the purpose of a folder?**
- **How do you rename a file?**
- **What is a shortcut?**
- **Name any four common OS utilities**

Thank You